Universal DSA notes

1 Analytical algorithm

The universal DSA scheme is given below. Assume $x \in [0, x_0]$ and that there are G groups. The S_N equations are below:

$$\mu \frac{\partial \psi_1^{\ell+\frac{1}{2}}}{\partial x} + \sigma_{t,1}\psi_1^{\ell+\frac{1}{2}} = \frac{1}{4\pi}\sigma_{1\to 1}\phi_1^{\ell}, \tag{1}$$

$$\mu \frac{\partial \psi_g^{\ell+\frac{1}{2}}}{\partial x} + \sigma_{t,g}\psi_g^{\ell+\frac{1}{2}} = \frac{1}{4\pi}\sigma_{g\to g}\phi_g^{\ell} + \frac{1}{4\pi}\sigma_{(g-1)\to g}\phi_{(g-1)}^{\ell}, \quad g = 2, 3, \ldots, G. \tag{2}$$

The boundary conditions are below:

$$\psi_g(0, \mu) = 0, \quad \mu > 0, \quad g = 1, 2, \ldots, G, \tag{3}$$

$$\psi_1(x_0, \mu) = 1, \quad \mu < 0, \tag{4}$$

$$\psi_g(x_0, \mu) = 0, \quad \mu < 0, \quad g = 2, 3, \ldots, G. \tag{5}$$

The diffusion acceleration step is given below:

$$-\frac{\partial}{\partial x}\frac{1}{3\sigma_{t,1}}\frac{\partial \phi_1^{\ell+1}}{\partial x} + \sigma_{R,1}\phi_1^{\ell+1} = -\frac{\partial}{\partial x}\left(J_1^{\ell+\frac{1}{2}} + \frac{1}{3\sigma_{t,1}}\frac{\partial \phi_1^{\ell+\frac{1}{2}}}{\partial x}\right), \tag{6}$$

$$-\frac{\partial}{\partial x}\frac{1}{3\sigma_{t,g}}\frac{\partial \phi_g^{\ell+1}}{\partial x} + \sigma_{R,g}\phi_g^{\ell+1} = \sigma_{(g-1)\to g}\phi_{(g-1)}^{\ell+1} - \frac{\partial}{\partial x}\left(J_g^{\ell+\frac{1}{2}} + \frac{1}{3\sigma_{t,g}}\frac{\partial \phi_g^{\ell+\frac{1}{2}}}{\partial x}\right), \quad g = 2, 3, \ldots, G. \tag{7}$$

The boundary conditions of the diffusion acceleration step are below, where $J_g^{\ell+1} = -\frac{1}{3\sigma_{t,g}}\frac{\partial \phi_g^{\ell+1}}{\partial x} + J_g^{\ell+\frac{1}{2}} + \frac{1}{3\sigma_{t,g}}\frac{\partial \phi_g^{\ell+\frac{1}{2}}}{\partial x}$:

$$J_g^{\ell+1}(0) + \langle\mu\rangle\phi_g^{\ell+1}(0) = J_g^{\ell+\frac{1}{2}}(0) + \langle\mu\rangle\phi_g^{\ell+\frac{1}{2}}(0), \quad g = 1, 2, \ldots, G, \tag{8}$$

$$J_g^{\ell+1}(x_0) - \langle\mu\rangle\phi_g^{\ell+1}(x_0) = J_g^{\ell+\frac{1}{2}}(x_0) - \langle\mu\rangle\phi_g^{\ell+\frac{1}{2}}(x_0), \quad g = 1, 2, \ldots, G, \tag{9}$$

where $\langle\mu\rangle = \frac{1}{2\pi}\sum_{\mu_m>0}\mu_m w_m$ is computed with the S_N quadrature, whose weights sum to 4π. These are Marshak conditions with no incoming current for $\phi_g^{\ell+1} - \phi_g^{\ell+\frac{1}{2}}$. The incident flux of group 1 enters through $J_1^{\ell+\frac{1}{2}}$ and $\phi_1^{\ell+\frac{1}{2}}$, so there is no separate boundary source. The cross-sections are defined as follows:

$$\sigma_{t,g} = \sigma_{a,g} + \sigma_{g\to g} + \sigma_{g\to(g+1)}, \quad g = 1, 2, \ldots, G-1, \tag{10}$$

$$\sigma_{R,g} = \sigma_{a,g} + \sigma_{g\to(g+1)}, \quad g = 1, 2, \ldots, G-1, \tag{11}$$

$$\sigma_{t,G} = \sigma_{a,G} + \sigma_{G\to G}, \tag{12}$$

$$\sigma_{R,G} = \sigma_{a,G}. \tag{13}$$

We can just use diamond-differencing for both the S_N and diffusion equations (look back at the NUEN 625 notes). We can also define a routine to define the cross-sections.

For the time being, take $x_0 = 10$ cm and the following as the cross-sections for all groups $g = 1, 2, \ldots, G$:

$$\sigma_{a,g} = 0.25\,\mathrm{cm}^{-1}, \tag{14}$$

$$\sigma_{g\to g} = 1.0\,\mathrm{cm}^{-1}, \tag{15}$$

$$\sigma_{g\to(g+1)} = 1.0\,\mathrm{cm}^{-1}. \tag{16}$$